ON THE INRADIUS OF POLYNOMIAL LEMNISCATES: THE NON-DEGENERATE CASE AND BEYOND

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Abstract. We prove that for any monic polynomial P of degree n with all roots in the closed unit disc $\overline{\mathbb{D}}$, if every critical value of P has modulus at least 1, then the lemniscate $\{z \in \mathbb{C} : |P(z)| < 1\}$ has inradius at least $1/(4n)$. This confirms the Erdős–Herzog–Piranian conjecture ($\rho_n \gg 1/n$) for the natural class of *non-degenerate* polynomials, which includes the conjectured extremizer $P(z) = z^n - 1$. The proof combines Koebe's quarter theorem with a Fekete energy selection argument and is independent of the recent area-based approach of Krishnapur–Lundberg–Ramachandran (2025), who applied a Nazarov–Polterovich–Sodin sublevel set bound to establish $\rho_n \geq c/(n\sqrt{\log n})$ for all polynomials. We further analyze the squarefree degenerate case via Blaschke product factorizations, identify an obstruction to extending the conformal method directly to merged components, and show that the full conjecture would follow if an inradius minimizer were non-degenerate. Preliminary numerics are consistent with this scenario.

1. Introduction

For a monic polynomial $P(z) = \prod_{i=1}^{n}(z - z_i)$ with coefficients in $\mathbb{C}$, the *lemniscate* at level $t > 0$ is

$$\Lambda_P(t) \;=\; \big\{z \in \mathbb{C} : |P(z)| < t\big\}.$$

When all roots satisfy $|z_i| \leq 1$, we write $P \in \mathcal{P}_n(\overline{\mathbb{D}})$. The *inradius* of a bounded open set $\Omega \subset \mathbb{C}$ is

$$\operatorname{inrad}(\Omega) \;=\; \sup\big\{r > 0 : B(z, r) \subset \Omega \text{ for some } z \in \mathbb{C}\big\}.$$

In their foundational 1958 paper [2], Erdős, Herzog, and Piranian posed the problem of determining the minimal inradius

$$\rho_n \;=\; \inf\big\{\operatorname{inrad}(\Lambda_P(1)) : P \in \mathcal{P}_n(\overline{\mathbb{D}})\big\}.$$

The polynomial $P(z) = z^n - 1$ has n roots at the n-th roots of unity and yields $\operatorname{inrad}(\Lambda_{z^n-1}) \sim \pi/(2n)$, suggesting the conjecture:

Conjecture 1.1 (Erdős–Herzog–Piranian). *$\rho_n \asymp 1/n$.*

Pommerenke [4] established the first lower bound $\rho_n \geq 1/(2en^2)$ in 1961, which remained the state of the art for 64 years. In a remarkable 2025 breakthrough, Krishnapur, Lundberg, and Ramachandran [3] proved

$$\rho_n \;\geq\; \frac{c}{n\sqrt{\log n}}, \tag{1}$$

confirming the conjecture up to a factor of $\sqrt{\log n}$. Their proof applies a Nazarov–Polterovich–Sodin lower bound on the area of harmonic sublevel sets with an isoperimetric inequality relating area, perimeter, and inradius of lemniscate components. The $\sqrt{\log n}$ loss arises from the square root in the area-to-inradius conversion. (Separately, Tao [5] recently resolved the related *length* conjecture from the same 1958 paper, establishing that the maximal arclength of $\{|P(z)| = 1\}$ is $\Theta(n)$ for sufficiently large n.)

In this note, we take a different approach. We identify a natural class of polynomials—those whose critical values all lie outside the open unit disc—for which the sharp $1/n$ bound holds. Our proof bypasses the area estimates entirely, using instead the univalent branch structure of P together with a classical energy bound. We then analyze the degenerate case (where some

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critical values lie inside $\mathbb{D}$) via Blaschke product factorizations, identify a structural obstruction that prevents naïve extensions of the conformal method, and show that the full conjecture would follow if some inradius minimizer were non-degenerate.

2. Definitions and statement of results

Definition 2.1. A monic polynomial $P \in \mathcal{P}_n(\overline{\mathbb{D}})$ is called *non-degenerate* if every critical value of P has modulus at least 1:

$$P'(\zeta) = 0 \quad \Longrightarrow \quad |P(\zeta)| \geq 1.$$

Let $\mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$ denote the set of non-degenerate polynomials in $\mathcal{P}_n(\overline{\mathbb{D}})$.

The non-degeneracy condition means that the polynomial P, viewed as a branched covering $P\colon \mathbb{C} \to \mathbb{C}$, has no branch points over the open unit disc $\mathbb{D}$. Equivalently, every branch of P^{-1} extends univalently from 0 to the boundary $|w| = 1$.

Example 2.2. The polynomial $P(z) = z^n - 1$ has a unique critical point at $\zeta = 0$, with critical value $P(0) = -1$ and $|P(0)| = 1$. Thus $z^n - 1 \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$.

Example 2.3. More generally, if P has n distinct roots and all roots lie on $\mathbb{T}$, then $P \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$ whenever $|P(\zeta)| \geq 1$ for each of the $n-1$ critical points. Numerical evidence suggests this holds for a wide class of polynomials with roots on or near the unit circle.

Our main result is:

Theorem 2.4 (Inradius bound for non-degenerate lemniscates)**.** *Let $P \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$. Then*

$$\operatorname{inrad}(\Lambda_P(1)) \;\geq\; \frac{1}{4n}.$$

Since $z^n - 1$ is non-degenerate with $\operatorname{inrad}(\Lambda_{z^n-1}) \sim \pi/(2n)$ (Proposition 4.1), the bound has the sharp order $1/n$.

Remark 2.5. The constant $1/4$ arises from the Koebe quarter theorem and is not optimal. For the extremal polynomial $z^n - 1$ the true constant is $\pi/2 \approx 1.571$. The Koebe bound is sharp for the full class of univalent functions, but the inverse branches ψ_j satisfy additional constraints (e.g., $\psi_j(\mathbb{D})$ is a petal of a polynomial lemniscate). Whether refined distortion theorems or Bieberbach-type coefficient bounds for ψ_j can improve the constant for $\mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$ remains an interesting question.

As an immediate consequence, we recover the EHP conjecture for the non-degenerate class:

Corollary 2.6. $\inf\bigl\{\operatorname{inrad}(\Lambda_P(1)) : P \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})\bigr\} \asymp 1/n.$

3. Proof of Theorem 2.4

The proof has three components: univalent branch extraction, the Koebe quarter theorem, and a Fekete energy argument for root selection.

3.1. Univalent inverse branches.

Lemma 3.1. *Let $P \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$ with distinct roots $z_1, \dots, z_n$. For each $j = 1, \dots, n$, there exists a holomorphic function $\psi_j\colon \mathbb{D} \to \mathbb{C}$ such that:*

(i) $P(\psi_j(w)) = w$ for all $w \in \mathbb{D}$;
(ii) $\psi_j(0) = z_j$;
(iii) ψ_j is univalent on $\mathbb{D}$;
(iv) $\psi_j(\mathbb{D}) \subset \Lambda_P(1)$.

Proof. Since $P(z_j) = 0$ and $P'(z_j) \neq 0$ (non-degeneracy implies simple roots by Remark 3.2), the inverse function theorem provides a local holomorphic branch ψ_j near $w = 0$ with $\psi_j(0) = z_j$.

The branch ψ_j can be continued analytically along any path in $\mathbb{D}$ that avoids the critical values of P. By the non-degeneracy hypothesis, all critical values have modulus ≥ 1 and therefore lie outside $\mathbb{D}$. By the monodromy theorem, ψ_j extends to a single-valued holomorphic function on all of $\mathbb{D}$.

Univalence follows because $\psi_j(w_1) = \psi_j(w_2)$ would imply $w_1 = P(\psi_j(w_1)) = P(\psi_j(w_2)) = w_2$.

Finally, $w \in \mathbb{D}$ implies $|P(\psi_j(w))| = |w| < 1$, so $\psi_j(w) \in \Lambda_P(1)$. □

Remark 3.2. If P has a repeated root at z_j of multiplicity $m \geq 2$, then z_j is itself a critical point with critical value $P(z_j) = 0$. The non-degeneracy condition then requires $|P(z_j)| \geq 1$, but $P(z_j) = 0$, a contradiction. Therefore, all roots of a non-degenerate polynomial are simple.

3.2. **The Koebe quarter theorem.**
We recall the classical result (see e.g. [1, Chapter 2]):

Theorem 3.3 (Koebe, 1907)**.** *If $f\colon \mathbb{D} \to \mathbb{C}$ is univalent with $f(0) = a$ and $f'(0) = b$, then*

$$f(\mathbb{D}) \supset B\Big(a, \tfrac{|b|}{4}\Big).$$

Applying Theorem 3.3 to ψ_j from Lemma 3.1:

Corollary 3.4. *For each root z_j,*

$$\Lambda_P(1) \supset \psi_j(\mathbb{D}) \supset B\left(z_j, \frac{1}{4|P'(z_j)|}\right).$$

Proof. We have $\psi_j(0) = z_j$ and $\psi_j'(0) = 1/P'(z_j)$, the latter following from differentiating $P(\psi_j(w)) = w$ at $w = 0$. □

3.3. **Root selection via Fekete energy.**
To complete the proof, we must show that at least one root has $|P'(z_j)| \leq n$.

Lemma 3.5 (Fekete energy bound)**.** *Let $z_1, \ldots, z_n \in \overline{\mathbb{D}}$ be distinct. Then*

$$\min_{1 \leq j \leq n} |P'(z_j)| \leq n.$$

Proof. Since $P(z) = \prod_{i=1}^n (z - z_i)$, we have $P'(z_j) = \prod_{i \neq j}(z_j - z_i)$, and therefore

$$\prod_{j=1}^n |P'(z_j)| = \prod_{j=1}^n \prod_{i \neq j} |z_j - z_i| = \Big(\prod_{1 \leq i < j \leq n} |z_i - z_j|\Big)^2.$$

Consider the Vandermonde matrix $V = (z_i^{k-1})_{1 \leq i,k \leq n}$, so that $|\det V| = \prod_{1 \leq i < j \leq n} |z_j - z_i|$. By Hadamard's inequality, $|\det V| \leq \prod_{i=1}^n \|r_i\|_2$, where $r_i = (1, z_i, \ldots, z_i^{n-1})$. Since $|z_i| \leq 1$,

$$\|r_i\|_2^2 = \sum_{k=0}^{n-1} |z_i|^{2k} \leq n,$$

so $|\det V| \leq n^{n/2}$. Therefore

$$\prod_{j=1}^n |P'(z_j)| \leq n^n.$$

The geometric mean of the $|P'(z_j)|$ is at most n, so $\min_j |P'(z_j)| \leq n$.

Equality is attained for the n-th roots of unity, where $|P'(\omega_k)| = n$ for every k, confirming sharpness. □

3.4. **Completion of the proof.**

Proof of Theorem 2.4. Let $P \in \mathcal{P}_n^{\mathrm{nd}}(\overline{\mathbb{D}})$. By Remark 3.2, all roots are simple. Choose j^* attaining $\min_j |P'(z_j)|$. By Lemma 3.5, $|P'(z_{j^*})| \leq n$. By Corollary 3.4,

$$\Lambda_P(1) \supset B\left(z_{j^*}, \frac{1}{4|P'(z_{j^*})|}\right) \supset B\left(z_{j^*}, \frac{1}{4n}\right).$$

Therefore $\mathrm{inrad}(\Lambda_P(1)) \geq 1/(4n)$. □

4. Verification on the extremal polynomial

Proposition 4.1. *For $P_n(z) = z^n - 1$, one has*

$$\operatorname{inrad}(\Lambda_{P_n}(1)) \sim \frac{\pi}{2n} \qquad \text{as } n \to \infty.$$

More precisely,

$$\liminf_{n\to\infty} n \cdot \operatorname{inrad}(\Lambda_{P_n}(1)) \geq \frac{\pi}{2} \qquad \text{and} \qquad \operatorname{inrad}(\Lambda_{P_n}(1)) \leq \frac{\pi}{2n} + O\left(\frac{1}{n^2}\right).$$

In particular, the bound of Theorem 2.4 has the sharp order $1/n$.

Proof. By rotational symmetry, $\Lambda_{P_n}(1)$ is the union of n congruent components, so it suffices to analyze the component U_n containing 1. On U_n we may use the principal logarithm and write $z = e^{\xi/n}$. Then

$$z \in U_n \iff |e^{\xi} - 1| < 1 \text{ and } |\operatorname{Im} \xi| < \pi/2.$$

Hence $U_n = \exp(\Omega/n)$, where

$$\Omega := \{\xi = x + iy : x < \log(2\cos y),\ |y| < \pi/2\}.$$

We first show that $\operatorname{inrad}(\Omega) = \pi/2$. Since $\Omega \subset \{|y| < \pi/2\}$, every disc contained in Ω has radius at most $\pi/2$, so $\operatorname{inrad}(\Omega) \leq \pi/2$. Conversely, let $0 < r < \pi/2$. Choose $M > r - \log(2\cos r)$. If $|\xi + M| < r$, then $|\operatorname{Im} \xi| < r$ and

$$\operatorname{Re} \xi \leq -M + r < \log(2\cos r) \leq \log(2\cos(\operatorname{Im} \xi)),$$

so $B(-M, r) \subset \Omega$. Therefore $\operatorname{inrad}(\Omega) \geq r$ for every $r < \pi/2$, proving $\operatorname{inrad}(\Omega) = \pi/2$.

Lower bound on $\operatorname{inrad}(U_n)$. Fix $0 < r < \pi/2$ and choose M as above so that $B(-M, r) \subset \Omega$. Set $a_n = e^{-M/n}$. For $|h| \leq r$,

$$e^{(-M+h)/n} = a_n\left(1 + \frac{h}{n} + R_n(h)\right),$$

where $|R_n(h)| \leq C_r/n^2$ for some constant C_r depending only on r. Fix $\varepsilon \in (0, r)$. If $|w - a_n| < a_n(r - \varepsilon)/n$, set

$$F(h) = a_n \frac{h}{n} - (w - a_n), \qquad G_n(h) = a_n R_n(h).$$

On $|h| = r$ we have $|F(h)| \geq a_n \varepsilon/n$ and $|G_n(h)| \leq a_n C_r/n^2$. Hence for all sufficiently large n, $|G_n(h)| < |F(h)|$ on $|h| = r$. By Rouché's theorem, $F + G_n$ has a zero in $|h| < r$, i.e. there exists $|h| < r$ such that $e^{(-M+h)/n} = w$. Therefore

$$B\left(a_n,\, a_n \frac{r - \varepsilon}{n}\right) \subset U_n$$

for all large n. Since $a_n = 1 + O(1/n)$, $\liminf_{n\to\infty} n \cdot \operatorname{inrad}(U_n) \geq r - \varepsilon$. Letting $\varepsilon \downarrow 0$ and then $r \uparrow \pi/2$,

$$\liminf_{n\to\infty} n \cdot \operatorname{inrad}(U_n) \geq \frac{\pi}{2}.$$

Upper bound on $\operatorname{inrad}(U_n)$. If $z \in U_n$, then $|z|^n < 2$ and $|\arg z| < \pi/(2n)$. Thus U_n is contained in the sector $\{z : |\arg z| < \pi/(2n)\}$ and in $\{|z| < 2^{1/n}\}$. If $B(z_0, \rho) \subset U_n$, then ρ is at most the distance from z_0 to the boundary of the sector, hence

$$\rho \leq |z_0| \sin\left(\frac{\pi}{2n}\right) \leq 2^{1/n} \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{2n} + O\left(\frac{1}{n^2}\right).$$

Combining the bounds gives $\operatorname{inrad}(U_n) \leq \pi/(2n) + O(1/n^2)$ and $\liminf_{n\to\infty} n \cdot \operatorname{inrad}(U_n) \geq \pi/2$, which together yield $\operatorname{inrad}(\Lambda_{P_n}(1)) \sim \pi/(2n)$. □

5. The non-degenerate class

The non-degeneracy condition $|P(\zeta)| \ge 1$ at all critical points has a clean potential-theoretic interpretation. The logarithmic potential $u(z) = \log|P(z)|$ satisfies $u(z_j) = -\infty$ at the roots and $u(\zeta) \ge 0$ at the critical points. Away from the roots, the zeros of P' are exactly the critical points of u; since u is harmonic there, simple critical points are of saddle type. Non-degeneracy means that u is non-negative at all its finite critical points.

Lemma 5.1. *P is non-degenerate if and only if $\Lambda_P(1)$ has exactly n connected components, each containing a single simple root.*

Proof. Every connected component U of $\Lambda_P(1)$ is simply connected (if U had a bounded complementary component H, then $\log|P|$ would be harmonic on H with boundary value 0, forcing $\log|P| \equiv 0$ on H, a contradiction). The restriction $P|_U \colon U \to \mathbb{D}$ is a proper holomorphic map whose degree equals the number of roots in U counted with multiplicity.

($\Rightarrow$) If P is non-degenerate, P has no critical values in $\mathbb{D}$, so $P|_U$ is an unbranched covering of $\mathbb{D}$. Since $\mathbb{D}$ is simply connected, this covering has degree 1: each component contains exactly one simple root, and there are n components.

($\Leftarrow$) If $\Lambda_P(1)$ has n components each with one root, then $P|_U$ has degree 1 on each component, hence no critical points in $\Lambda_P(1)$. Every critical value therefore has modulus ≥ 1. □

This structural property—that the lemniscate has n separated petals—is essential to our proof and is the geometric content of non-degeneracy.

6. Analysis of the degenerate case

When some critical value satisfies $|P(\zeta)| < 1$, the lemniscate components merge: two or more roots share a connected component. The univalent branch ψ_j can no longer be extended to all of $\mathbb{D}$, as it encounters a branch point at $w = P(\zeta)$.

We present here our analysis of the degenerate case, including both positive results and structural obstructions that clarify the difficulty of the full EHP conjecture.

Remark 6.1. In Sections 6–6.5 we restrict attention to squarefree polynomials (i.e., polynomials with simple roots) unless explicitly stated otherwise. This excludes repeated roots, for which the formulas involving $P'(z_j)$, Blaschke zeros, and critical-point expansions require separate treatment.

6.1. **The conformal factorization.**

Assume that P is squarefree. Let z_{j^*} be a root satisfying

$$|P'(z_{j^*})| \;=\; \min_{1\le j\le n} |P'(z_j)|.$$

By Lemma 3.5, $|P'(z_{j^*})| \le n$. Let $\mathcal{C}$ be the connected component of $\Lambda_P(1)$ containing z_{j^*}. Suppose $\mathcal{C}$ contains $k \ge 2$ roots (the degenerate case). Let $\varphi \colon \mathbb{D} \to \mathcal{C}$ be the Riemann map with $\varphi(0) = z_{j^*}$ (this exists since $\mathcal{C}$ is simply connected, as shown in the proof of Lemma 5.1). The composition $B = P \circ \varphi \colon \mathbb{D} \to \mathbb{D}$ is a degree-k Blaschke product with $B(0) = 0$.

By the chain rule, $\varphi'(0) = B'(0)/P'(z_{j^*})$, and the Koebe quarter theorem gives

$$\operatorname{inrad}(\mathcal{C}) \;\ge\; \frac{|B'(0)|}{4|P'(z_{j^*})|}. \tag{2}$$

Since $B(w) = w \prod_{i=2}^{k} \frac{w-\beta_i}{1-\bar{\beta}_i w}$ (the standard representation with $B(0) = 0$), evaluating the derivative at $w = 0$ via the product rule gives $|B'(0)| = \prod_{i=2}^{k} |\beta_i|$, where $\beta_2, \ldots, \beta_k$ are the φ-preimages of the other roots in $\mathcal{C}$. Since $|P'(z_{j^*})| \le n$,

$$\operatorname{inrad}(\mathcal{C}) \;\ge\; \frac{\prod_{i=2}^{k} |\beta_i|}{4n}.$$

We emphasize that this bound applies to the component containing the Fekete-selected root; for an arbitrary component, no such estimate is available from the present arguments.

For the degenerate case, the Blaschke zeros β_i may cluster near the origin, causing $\prod|\beta_i|$ to be small. To capture the "fatness" of the merged component, one may instead evaluate the conformal radius at a critical point of B.

6.2. **Conformal radius at critical points.** Let $w_c \in \mathbb{D}$ be a critical point of B (i.e., $B'(w_c) = 0$), and let $\zeta_c = \varphi(w_c)$ be the corresponding critical point of P. The Koebe bound at w_c gives

$$\operatorname{inrad}(\mathcal{C}) \;\geq\; \frac{1}{4}|\varphi'(w_c)|\,(1-|w_c|^2). \tag{3}$$

To express $|\varphi'(w_c)|$ in terms of the polynomial and Blaschke data, we use a local expansion.

Lemma 6.2. *Let $\varphi\colon \mathbb{D} \to \mathcal{C}$ be conformal and set $B = P \circ \varphi$. Let $w_c \in \mathbb{D}$ and $\zeta_c = \varphi(w_c)$. Then w_c is a critical point of B of multiplicity $m \geq 1$ if and only if ζ_c is a critical point of P of multiplicity m. In that case, $B^{(1)}(w_c) = \cdots = B^{(m)}(w_c) = 0$ and*

$$B^{(m+1)}(w_c) \;=\; P^{(m+1)}(\zeta_c)\,\varphi'(w_c)^{m+1}. \tag{4}$$

Proof. Write the local expansions $\varphi(w) = \zeta_c + a(w-w_c) + O((w-w_c)^2)$ with $a = \varphi'(w_c) \neq 0$, and $P(z) = P(\zeta_c) + c\,(z-\zeta_c)^{m+1} + O((z-\zeta_c)^{m+2})$ where $c = P^{(m+1)}(\zeta_c)/(m+1)!$. Substituting gives $B(w) = P(\zeta_c) + c\,a^{m+1}(w-w_c)^{m+1} + O((w-w_c)^{m+2})$, which is the claimed statement. $\square$

Corollary 6.3. *If w_c is a simple critical point of B, then $\zeta_c = \varphi(w_c)$ is a simple critical point of P and*

$$|\varphi'(w_c)|^2 \;=\; \frac{|B''(w_c)|}{|P''(\zeta_c)|}. \tag{5}$$

Proof. Apply Lemma 6.2 with $m = 1$. Then $B''(w_c) = P''(\zeta_c)\,\varphi'(w_c)^2$, and taking absolute values gives (5). $\square$

This cleanly separates the "microscopic" Blaschke geometry (the numerator) from the "macroscopic" polynomial scaling (the denominator).

6.3. **A structural obstruction for $k \geq 2$.** For $k = 2$, the Blaschke product $B(w) = w \cdot \frac{w-\beta}{1-\bar{\beta}w}$ has a unique critical point w_c inside $\mathbb{D}$. Direct computation gives

$$|B''(w_c)|(1-|w_c|^2)^2 \;=\; 2(1-|B(w_c)|^2). \tag{6}$$

As the two roots in the component move further apart and the lemniscate approaches a pinch point, the critical value satisfies $|B(w_c)| \to 1$, and consequently the hyperbolic second derivative degenerates to zero. Thus the conformal radius estimate (3) fails to provide a universal lower bound even for two-root components.

For $k \geq 3$, the same obstruction persists in a stronger form. Consider the family of degree-3 Blaschke products $B_a(w) = w^2 \cdot \frac{w-a}{1-aw}$ for $a \in (0,1)$. A direct computation gives

$$B_a'(w) \;=\; \frac{w\big(-2a + (3+a^2)w - 2aw^2\big)}{(1-aw)^2}.$$

The point $w_c = 0$ is a critical point with $B_a''(0) = -2a$. The nonzero critical point w_c' is the small root of the quadratic $-2a + (3+a^2)w - 2aw^2 = 0$; for small a, this satisfies $w_c' = 2a/3 + O(a^3)$. A direct evaluation then gives $|B_a''(w_c')|(1-|w_c'|^2)^2 = 2a + O(a^3)$.

As $a \to 0$, the hyperbolic second derivative at *both* critical points tends to zero:

$$\max_{w_c} |B_a''(w_c)|(1-|w_c|^2)^2 \;\to\; 0.$$

Consequently, the conformal radius estimate (3) degenerates, and the particular low-order Blaschke critical-point invariants considered above do not by themselves yield a universal bound for $k \geq 2$.

Remark 6.4. Geometrically, as $a \to 0$, the Blaschke product approaches $B_0(w) = w^3$. In the limit, the critical point at 0 has multiplicity 2, and Lemma 6.2 gives $|\varphi'(0)|^3 = |B'''(0)|/|P'''(\zeta_c)| = 6/|P'''(\zeta_c)|$, which is bounded away from zero. The obstruction is thus a *near-degeneracy* phenomenon: the Blaschke invariants degenerate continuously as critical points approach each other, even though they "reset" at the exact moment of collision via the higher derivative. This indicates that a proof for merged components cannot rely solely on a uniform lower bound for one distinguished low-order Blaschke derivative, though it does not rule out other conformal approaches.

6.4. **An identity for P'' at critical points.** Despite the Blaschke obstruction, the polynomial structure provides useful identities. Differentiating the logarithmic derivative $P'(z)/P(z) = \sum_{j=1}^{n} 1/(z - z_j)$ and evaluating at a critical point ζ where $P'(\zeta) = 0$ and $P(\zeta) \neq 0$ (i.e., ζ is not a repeated root), we obtain

$$P''(\zeta) \;=\; -P(\zeta) \sum_{j=1}^{n} \frac{1}{(\zeta - z_j)^2}. \tag{7}$$

This identity cleanly separates local and global contributions: roots near ζ dominate the sum via their large $1/(\zeta - z_j)^2$ terms, while distant roots contribute negligibly.

Lemma 6.5 (Dual Fekete bound)**.** *Let P be a monic polynomial of degree n with all roots in $\overline{\mathbb{D}}$. If P' has a repeated zero, then $\min_{\zeta:\, P'(\zeta)=0} |P''(\zeta)| = 0$. Otherwise, if $\zeta_1, \dots, \zeta_{n-1}$ are the distinct critical points of P, then*

$$\min_{1 \le i \le n-1} |P''(\zeta_i)| \;\le\; n(n-1). \tag{8}$$

Proof. If P' has a repeated zero, the first claim is immediate. Assume now that P' has distinct zeros $\zeta_1, \dots, \zeta_{n-1}$. Since $P'(z) = n \prod_{j=1}^{n-1} (z - \zeta_j)$, we have $P''(\zeta_i) = n \prod_{j \neq i} (\zeta_i - \zeta_j)$. By the Gauss–Lucas theorem, each ζ_i lies in $\overline{\mathbb{D}}$. Applying Lemma 3.5 to the $n-1$ points $\zeta_1, \dots, \zeta_{n-1}$ gives $\min_i \prod_{j \neq i} |\zeta_i - \zeta_j| \le n - 1$, and therefore $\min_i |P''(\zeta_i)| \le n(n-1)$. □

This "dual Fekete bound" establishes that the critical points are subject to the same macroscopic energy constraints as the roots.

6.5. **Numerical evidence.** We computed the inradius for $P \in \mathcal{P}_n(\overline{\mathbb{D}})$ with $n = 10$ across several hundred random configurations: roots uniformly distributed on $\mathbb{T}$, roots in the interior of $\mathbb{D}$, clustered configurations, and adversarial gradient searches. (The inradius was approximated by evaluating $|P|$ on a fine polar grid and performing a binary search on the radius of the largest inscribed disc.) In all cases tested, the minimum value of $n \cdot \mathrm{inrad}(\Lambda_P)$ was attained by roots of unity, with $n \cdot \mathrm{inrad}(\Lambda_{z^{10}-1}) \approx 1.26$.

Degenerate configurations (with roots in the interior) consistently exhibited *larger* inradii—typically $n \cdot \mathrm{inrad} \ge 4$. Even among non-degenerate configurations on $\mathbb{T}$, the roots of unity appeared extremal; perturbing any root away from the uniform spacing increased the inradius.

These observations are consistent with the hypothesis that the inradius minimizer is non-degenerate (and in fact equals $z^n - 1$), which would immediately imply the full EHP conjecture via Theorem 2.4. Since these computations play no role in the proofs, we record them only as motivation.

7. Comparison with KLR approach

The KLR proof of (1) proceeds by:

(1) Using the Nazarov–Polterovich–Sodin theorem to show $\mathrm{Area}(\Lambda_P) \ge c/\log n$;
(2) Bounding the perimeter via a Fryntov–Nazarov formula: $L(\Lambda_P) \le 4n\sqrt{\pi \cdot \mathrm{Area}(\Lambda_P)}$;
(3) Applying an isoperimetric inequality: $\mathrm{Area} \le 18\pi \cdot \mathrm{inrad} \cdot L$.

Combining these gives $\mathrm{inrad} \ge c\sqrt{\mathrm{Area}}/n \ge c/(n\sqrt{\log n})$. The $\sqrt{\log n}$ loss is structural: it arises from the square root in the area-to-inradius conversion via the perimeter bound.

Our approach is fundamentally different: we never estimate area or perimeter. Instead, we work directly with the *conformal structure* of the polynomial map $P\colon \mathbb{C} \to \mathbb{C}$, using the branched covering structure to construct large inscribed discs. This explains why we obtain the sharp $1/n$ order without logarithmic losses—the Koebe theorem provides a direct pointwise estimate, bypassing the global integral estimates that introduce the square root.

8. Reduction of the full conjecture

We summarize the implications of our analysis for the full EHP conjecture.

Lemma 8.1. *For each $n \geq 1$, there exists $P^* \in \mathcal{P}_n(\overline{\mathbb{D}})$ such that* $\operatorname{inrad}(\Lambda_{P^*}(1)) = \rho_n$.

Proof. The parameter space $\overline{\mathbb{D}}^n$ (identifying a monic polynomial with its n roots) is compact. It suffices to show that $a \mapsto \operatorname{inrad}(\Lambda_{P_a}(1))$ is continuous. All lemniscates are contained in $\{z : |z| \leq 2\}$ (since $|z| \geq 2$ and $|a_j| \leq 1$ imply $|P_a(z)| \geq 1$), so the inradius is bounded. Lower semicontinuity follows from uniform convergence of polynomials on compact sets: if $B(z_0, r) \subset \Lambda_{P_a}(1)$, then $\max_{\overline{B(z_0,r)}} |P_a| < 1$, and for a' sufficiently close to a, $|P_{a'}| < 1$ on $\overline{B(z_0, r)}$ as well. Upper semicontinuity follows similarly: if $|P_{a^{(m)}}| < 1$ on $B(z_m, r_m)$ with $z_m \to z$ and $r_m \to L$, then $|P_a| \leq 1$ on $B(z, r)$ for any $r < L$, hence $|P_a| < 1$ by the maximum principle (since P_a is nonconstant). A continuous function on a compact set attains its infimum. □

Proposition 8.2. *The following implications hold:*

(i) If some inradius minimizer in $\mathcal{P}_n(\overline{\mathbb{D}})$ is non-degenerate, then the EHP conjecture holds: $\rho_n \geq 1/(4n)$.

(ii) For every $P \in \mathcal{P}_n(\overline{\mathbb{D}})$, if there exists a connected component $\mathcal{C}$ of $\Lambda_P(1)$ containing k roots with $\operatorname{inrad}(\mathcal{C}) \geq c/n$, *then* $\operatorname{inrad}(\Lambda_P(1)) \geq c/n$.

In particular, one sufficient route to the full EHP conjecture is to prove that some inradius minimizer is non-degenerate.

Proof. For (i): if the minimizer P^* is non-degenerate, then Theorem 2.4 gives $\rho_n = \operatorname{inrad}(\Lambda_{P^*}) \geq 1/(4n)$. Statement (ii) is immediate since $\operatorname{inrad}(\Lambda_P) \geq \operatorname{inrad}(\mathcal{C})$. □

Remark 8.3. The converse of (i) does not hold a priori: the minimizer could theoretically be degenerate while still satisfying $\operatorname{inrad} \geq c/n$. However, no proof of non-degeneracy for the minimizer is currently known, and this remains a conjecture supported by the numerical evidence of Section 6.5.

Our analysis identifies two structural barriers to proving the full conjecture directly for degenerate components:

(i) The examples of Section 6.3 show that the particular low-order Blaschke critical-point invariants considered here can degenerate for $k \geq 2$, so this line of argument does not by itself yield a universal bound.

(ii) The inradius is not monotone under root perturbation (unlike area, which decreases under radial pushing by the zero-pushing lemma of [3]), preventing variational proofs of non-degeneracy for the minimizer.

We propose three possible routes to the full conjecture:

1. **Non-degeneracy of minimizers.** Prove that the inradius minimizer is non-degenerate by non-variational methods (e.g., direct analysis of the extremal conditions). The numerical evidence of Section 6.5 is consistent with this.
2. **Curvature-based width bounds.** The total curvature of a degree-n lemniscate boundary satisfies $\int |\kappa|\, ds \leq 2\pi n$. This prevents arbitrarily thin, wiggly components and may yield direct width bounds that bypass the area-perimeter chain.
3. **Capacity-diameter-inradius inequalities.** For a connected monic lemniscate $\{z : |P(z)| < 1\}$ of degree n with $\operatorname{Cap} = 1$ and all roots in $\overline{\mathbb{D}}$, prove $\operatorname{inrad} \geq c/n$ using potential-theoretic methods. The capacity constraint is the key rigidity that separates polynomial lemniscates from general harmonic sublevel sets (for which the analogous statement is false).

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